Sajt 4-8 oct 2021 — R. Iridescu

Exercițiu Să se stabilească natura pentru fiecare din următoarele serii numerice:

1) $\sum_{n\geq 0} \operatorname{tg}\left(\frac{\pi n}{4n+1}\right)$; 2) $\sum_{n\geq 0} \frac{1}{(4n+3)(4n+7)}$

3) $\sum_{n\geq 1} \frac{n^3\cdot(-2)^n + 5^{2n}\cdot n\sqrt{n}}{n^3\cdot 5^{2n+1}}$; 4) $\sum_{n\geq 1} \frac{(-1)^n\cdot n^4+n^2+1}{n^2(n^2+1)}$

5) $\sum_{n\geq 0} n \ln\left(\frac{2n+1}{2n+3}\right)$; 6) $\sum_{n\geq 2} \frac{1}{n\ln n}$;

7) $\sum_{n\geq 1} (n^2+n)\left[\frac{1}{n+1}\sin\frac{1}{n} - \frac{1}{n}\sin\frac{1}{n+1}\right]$

8) $\sum_{n\geq 1} n\left(2^{\frac{3n}{n^2+1}} - 3^{\frac{n^2}{n^3+1}}\right)$; 9) $\sum_{n\geq 1} \frac{e^n}{e^{2n}+1}$

10) $\sum_{n\geq 1} \left(1+\operatorname{tg}\frac{1}{n}\operatorname{tg}\frac{1}{n+1}\right)\operatorname{tg}\frac{1}{n(n+1)}$

Sol Vom nota cu x_n termenul general al fiecărei serii

1) $\lim_{n\to\infty} x_n = \operatorname{tg}\frac{\pi}{4} = 1 \neq 0 \overset{\text{Cril necesar}}{\Longrightarrow} \sum_{n\geq 0} \operatorname{tg}\left(\frac{\pi n}{4n+1}\right)$ (SD)

2) $x_n = \frac{1}{4}\cdot\left(\frac{1}{4n+3} - \frac{1}{4n+7}\right) \Rightarrow S_n = \sum_{k=0}^{n} x_k =$

$= \frac{1}{4}\left(\sum_{k=0}^{n}\frac{1}{4k+3} - \sum_{k=0}^{n}\frac{1}{4k+7}\right) = \frac{1}{4}\left[\frac{1}{3}+\frac{1}{7}+\frac{1}{11}+\ldots+\right.$

$\left.+\frac{1}{4n+3} - \left(\frac{1}{7}+\frac{1}{11}+\ldots+\frac{1}{4n+3}+\frac{1}{4n+7}\right)\right] = \frac{1}{4}\left(\frac{1}{3}-\frac{1}{4n+7}\right)$

$\xrightarrow{n\to\infty} \frac{1}{12} \in \mathbb{R} \overset{\text{def}}{\Rightarrow} \sum_{n\geq 0} \frac{1}{(4n+3)(4n+7)}$ (SC) și are suma

$\sum_{n=0}^{\infty} \frac{1}{(4n+3)(4n+7)} = \frac{1}{12}$ ∎

3) $\sum_{n\geq 1} x_n = \frac{1}{5} \cdot \sum_{n\geq 1} \left(-\frac{2}{25}\right)^n + \frac{1}{5} \sum_{n\geq 1} \frac{1}{n\sqrt{n}}$

(SC) (SC) $\Rightarrow$

(serie geometrică cu rația $q = -\frac{2}{25} \in (-1,1)$) (serie Riemann cu $\alpha = \frac{3}{2} > 1$)

$\Rightarrow \sum_{n\geq 1} x_n$ (SC) ∎

4) $\sum_{n\geq 1} x_n = \sum_{n\geq 1} (-1)^n \cdot \frac{n^2}{n^2+1} + \sum_{n\geq 1} \frac{1}{n^2}$

$\lim_{n\to\infty} (-1)^n \cdot \frac{n^2}{n^2+1}$ nu există $\overset{\text{Crit nec}}{\Rightarrow} \sum_{n\geq 1} (-1)^n \cdot \frac{n^2}{n^2+1}$ (SD)

$\sum_{n\geq 1} \frac{1}{n^2}$ (SC) (serie Riemann cu $\alpha = 2 > 1$)

$\Rightarrow$

$\Rightarrow \sum_{n\geq 1} x_n$ (SD) ∎

5) $x_n = \ln\left(\frac{2n+1}{2n+3}\right)^n = \ln\left(1 - \frac{2}{2n+3}\right)^n \xrightarrow[n\to\infty]{} \ln(e^{-1}) =$

$= -1 \neq 0 \overset{\text{Crit nec.}}{\Rightarrow} \sum_{n\geq 0} x_n$ (SD)

6) Fie $f: (2,+\infty) \to \mathbb{R}_+$, $f(x) = \frac{1}{x \ln x}$

Atunci: $f'(x) = -\frac{1}{x^2 \ln x} - \frac{1}{x^2 \ln^2 x} < 0 \Rightarrow f$ nu descrescătoare. Putem aplica criteriul integral:

$$\int_2^n f(x)\,dx = \int_2^n \frac{1}{x \ln x}\,dx = \ln(\ln x)\Big|_2^n = \ln\left(\frac{\ln n}{\ln 2}\right) \to$$

$\xrightarrow{n\to\infty} \infty \Rightarrow \lim_{n\to\infty}\left(\int_2^n f(x)dx\right)_{n\geq 2}$ divergent $\Rightarrow$ Crit. integral

$$\Rightarrow \sum_{n\geq 2} \underbrace{f(n)}_{\| \; x_n} \quad (SD) \blacksquare$$

7) $x_n = n\sin\frac{1}{n} - (n+1)\sin\frac{1}{n+1} \Rightarrow S_n = \sum_{k=1}^{n} x_k =$

$$= \sum_{k=1}^{n} k\sin\frac{1}{k} - \sum_{k=1}^{n}(k+1)\sin\frac{1}{k+1} = \sin 1 + 2\sin\frac{1}{2} + \cdots +$$

$$+ n\sin\frac{1}{n} - \left(2\sin\frac{1}{2} + 3\sin\frac{1}{3} + \cdots + n\sin\frac{1}{n} + (n+1)\sin\frac{1}{n+1}\right) =$$

$$= \sin 1 - (n+1)\sin\frac{1}{n+1} \xrightarrow{n\to\infty} (\sin 1) - 1 \Rightarrow \sum_{n\geq 1} x_n \; (SC) \text{ si}$$

$$\sum_{n=1}^{\infty} x_n = (\sin 1) - 1 \blacksquare$$

8) $$\lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{2^{\frac{3n}{n^2+1}} - 1}{\frac{1}{n}} - \frac{3^{\frac{n^2}{n^3+1}} - 1}{\frac{1}{n}} =$$

$$= \lim_{n\to\infty} \frac{2^{\frac{3n}{n^2+1}} - 1}{\frac{3n}{n^2+1}} \cdot \frac{3n^2}{n^2+1} - \frac{3^{\frac{n^2}{n^3+1}} - 1}{\frac{n^2}{n^3+1}} \cdot \frac{n^3}{n^3+1} =$$

$$= 3\ln 2 - \ln 3 = \ln\frac{8}{3} \neq 0 \overset{\text{Crit nec.}}{\Rightarrow} \sum_{n\geq 1} x_n \; (SD)$$

9) Fie $f : [1, +\infty) \to \mathbb{R}_+$, $f(x) = \dfrac{e^x}{e^{2x}+1}$.

Atunci: $f'(x) = \dfrac{e^x(e^{2x}+1) - 2e^{3x}}{(e^{2x}+1)^2} = \dfrac{e^x(1-e^{2x})}{(e^{2x}+1)^2} < 0 \Rightarrow$ $\forall x \geq 1$

$\Rightarrow f$ nu mai descr. $\Rightarrow$ putem aplica criteriul integral:

$$\int_1^n f(x)\,dx = \int_1^n \frac{e^x}{e^{2x}+1}\,dx \overset{t=e^x}{=} \int_e^{e^n} \frac{1}{t^2+1}\,dt =$$

$$= \operatorname{arctg}(e^n) - \operatorname{arctg} e \xrightarrow{n\to\infty} \frac{\pi}{2} - \operatorname{arctg} e \Rightarrow \text{şirul}$$

$\left(\int_1^n f(x)\,dx\right)_{n\geq 1}$ este convergent $\overset{\text{crit integral}}{\Rightarrow} \sum_{n\geq 1} \underbrace{f(n)}_{x_n}$ (sc) ■

10) $x_n = \left(1 + \operatorname{tg}\frac{1}{n}\operatorname{tg}\frac{1}{n+1}\right)\operatorname{tg}\left(\frac{1}{n} - \frac{1}{n+1}\right) = \operatorname{tg}\frac{1}{n} - \operatorname{tg}\frac{1}{n+1}$

$$\Rightarrow S_n = \sum_{k=1}^n x_k = \sum_{k=1}^n \operatorname{tg}\frac{1}{k} - \sum_{k=1}^n \operatorname{tg}\frac{1}{k+1} =$$

$$= \operatorname{tg} 1 + \operatorname{tg}\frac{1}{2} + \cdots + \operatorname{tg}\frac{1}{n} - \left(\operatorname{tg}\frac{1}{2} + \cdots + \operatorname{tg}\frac{1}{n} + \operatorname{tg}\frac{1}{n+1}\right) =$$

$$= \operatorname{tg} 1 - \operatorname{tg}\frac{1}{n+1} \xrightarrow{n\to\infty} \operatorname{tg} 1 \in \mathbb{R} \Rightarrow \text{şirul } (S_n)_{n\geq 1} \text{ este}$$

convergent $\Rightarrow \sum_{n\geq 1} x_n$ (sc) ■